

Exoplanets

Exercise 1

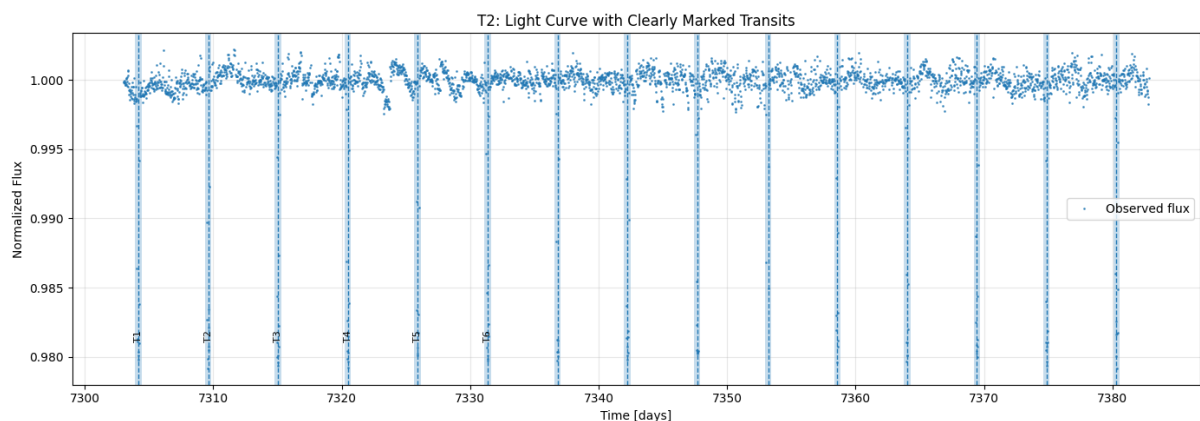
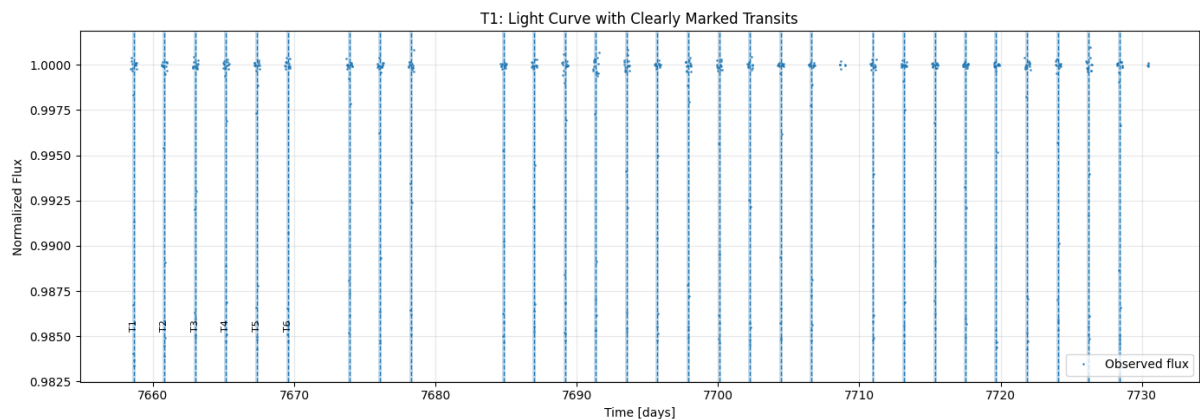
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Task 1.1: Plotting the light curve

The light curve was plotted with time on the x-axis and normalized flux on the y-axis. Most of the flux remains close to unity, representing the unobscured brightness of the star. However, periodic drops in flux are visible at regular intervals. These dips indicate the presence of a transiting object passing in front of the host star and blocking a fraction of its light.

Task 1.2: Finding the transits

The transits were identified as repeated decreases in flux below the baseline stellar brightness. Several of these dips were marked on the light curve. Their regular spacing and similar depth strongly suggest an orbiting body rather than random instrumental noise. These marked transits were later used to estimate the orbital period and epoch.



Task 1.3: First estimate of orbital period

The orbital period was estimated by measuring the time interval between consecutive transits. Since the same object returns once every orbit, the spacing between neighbouring transit centres provides the first approximation of the orbital period.

For T1:

$$P \approx 2.18 \text{ days}$$

For T2:

$$P \approx 5.45 \text{ days}$$

This estimate is preliminary because small uncertainties in locating the transit centres can affect the measured value.

Task 1.4: Estimate of mid-transit epoch

The epoch was estimated by determining the midpoint of one clear transit event. This corresponds to the deepest and most symmetric part of the transit and is defined as the mid-transit epoch, E .

For T1:

$$E \approx 7658.65 \text{ days}$$

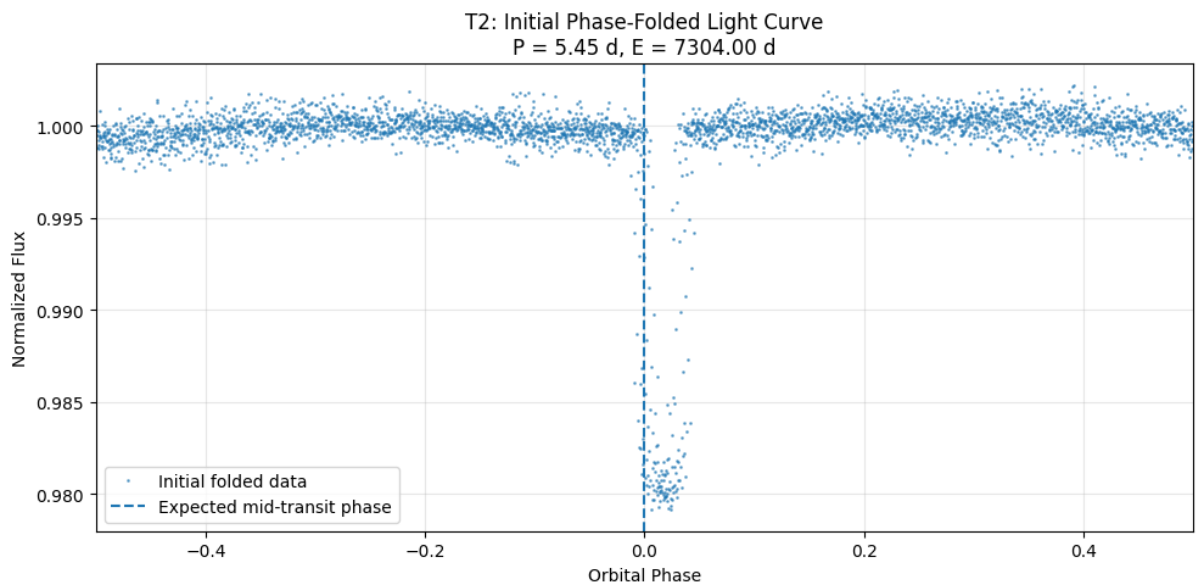
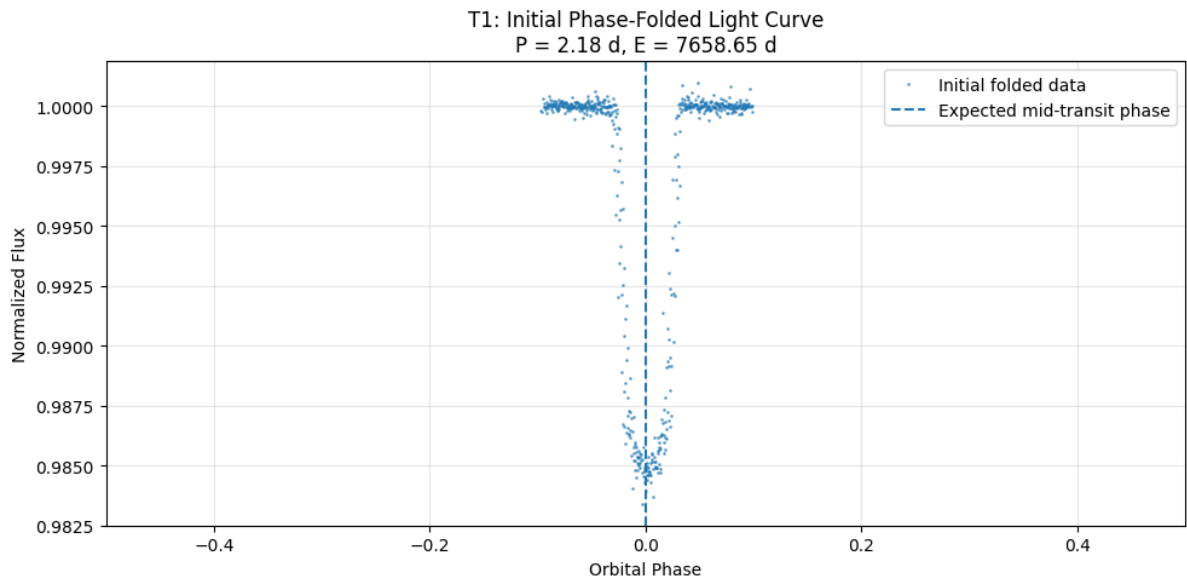
For T2:

$$E \approx 7304 \text{ days}$$

The epoch serves as the reference time used later for phase-folding the light curve.

Task 1.5: First phase-folded light curve

Using the initial period and epoch estimates, the light curve was phase-folded so that all transits overlap within a single orbital cycle. The transit feature appears near phase 0, although the alignment is not perfect due to small uncertainties in the initial period estimate. Even minor period errors accumulate over multiple orbits and broaden the transit in phase space.



Task 1.6: Improved period, epoch, and transit shape

The period and epoch were refined through trial-and-error adjustments until the transits aligned more accurately at phase 0. These refined values form the ephemeris of the system.

For T1:

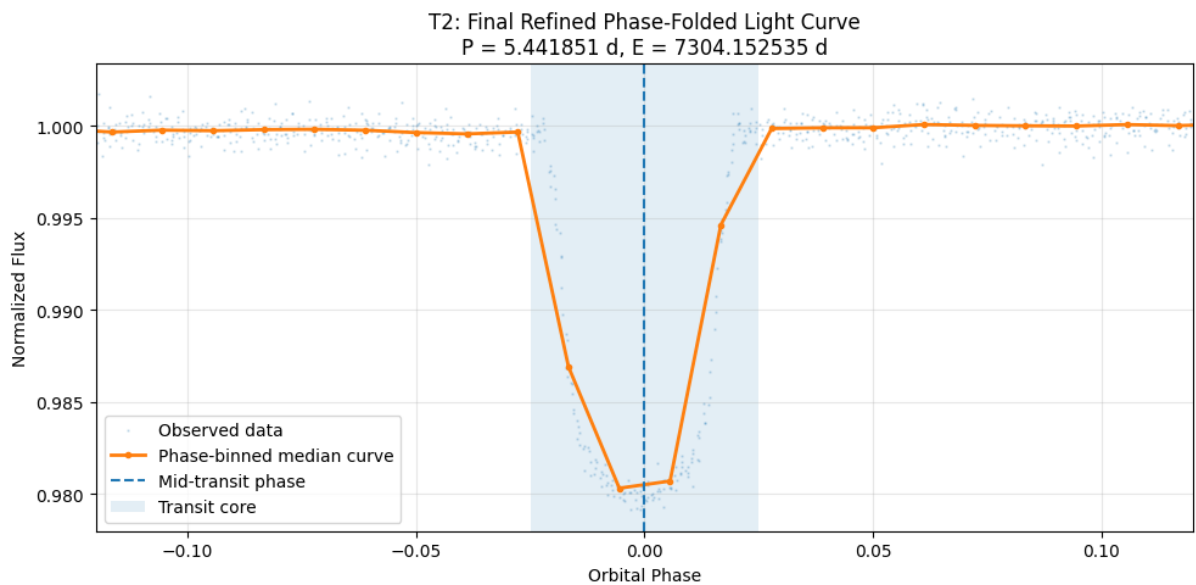
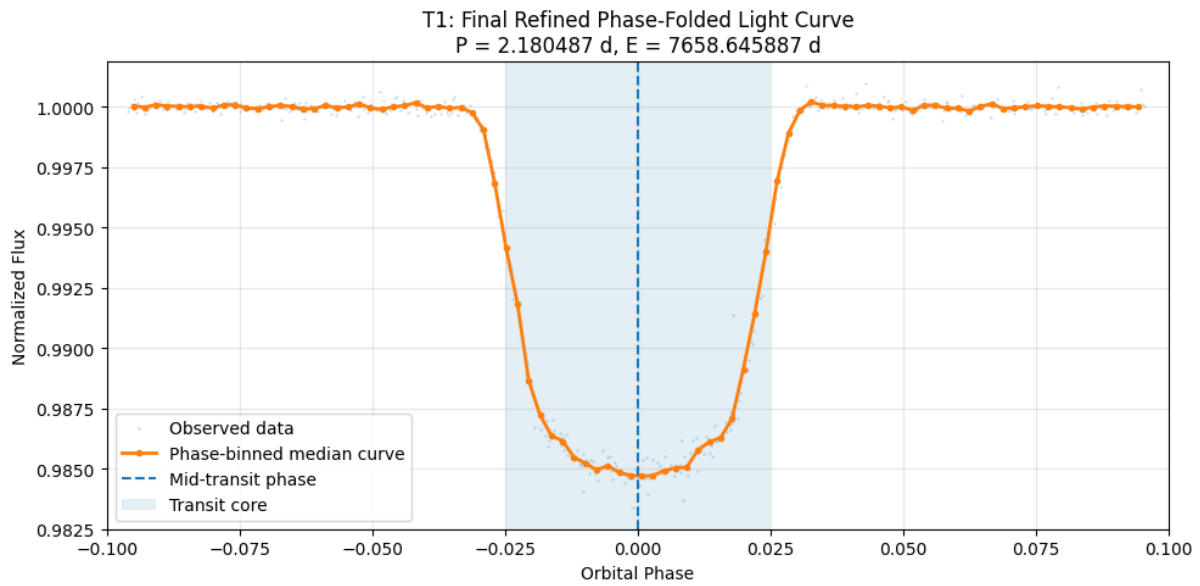
P = 2.1805 days

E = 5658.6459 days

For T2:

P = 5.4418 days

E = 7304.1525 days



The final transit profile is not perfectly trapezoidal because in real stars the stellar centre appears brighter than the edge. As a result, ingress and egress are smoother than in an idealized transit model. Instrumental noise and finite exposure time also contribute to deviations from a perfect trapezium.